MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 1DP1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Miniature 1-inch, electrostatic deflection and focus.

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Rg	Zd	Rd	Alt
Unit:	V	Vdc	v	Vdc	Vdc	Meg	Meg	Meg	ft
Maximum:	6.9	0, -100	400	300	1,000	2	2	2	60,000
Minimum:	5.7	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	Adjust	---	Focus	800	---	---	---	---

GENERAL:

(F) First article inspection - Required (see note 4)

(F) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5106	Bulb, screen, and face-plate quality	-		---	---	---	---
5223	Modulation	-	Light = 5 ftL	ΔEc	---	40	Vdc
5231	Spot position (electro-static deflection)	-		Displ	---	4	mm
5241	Grid cutoff voltage	-		Ec1	---	-95	Vdc
5251	Grid No. 1 leakage current	-		---	---	5	μAdc
5251	Anode No. 2 leakage current	-		---	---	15	μAdc
5206	Gas "cross"	1	Light = 5 ftL	---	---	---	---
5231	Spot displacement (leakage)	-		Displ	---	3	mm
5221	Light output	2		Light	5	---	ftL
	Quality conformance inspection, part 2						
1301	Heater current	-		If	180	220	mAdc
5216	Stray light emission (conventional types)	-	Eb2 = 1,000 Vdc	---	---	---	---
5226	Line width A (electrostatic deflection)	-	Ik = 50 μAdc, 12 lines	Width	---	0.25	mm
5226	Line width B (electrostatic deflection)	-	Ik = 50 μAdc, 12 lines	Width	---	0.35	mm
5201	Electrode current (anode No. 1)	-		Ib1	-10	+10	μAdc
5201	Electrode current (cathode)	-	Light = 5 ftL	Ik	---	1	mA
5248	Deflection factor	-	1D2	DF	350	450	Vdc in.
5248	Deflection factor	-	3D4	DF	250	350	Vdc in.
5101	Angle between traces	-		---	---	---	---
5246	Focusing voltage at cutoff	-		Fb1	175	275	Vdc
5101	Base alinement (electrostatic types)	-	1D2 between pin numbers 2 and 3	---	---	---	---
1105	Permanence of marking	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Eb2 = 800 Vdc; t = 500 hours minimum; light = 5 ftL	---	---	---	---
---	Life-test end points:						
5251	Grid No. 1 leakage current	-		---	---	5	µAdc
5251	Anode No. 2 leakage current	-		---	---	15	µAdc
5216	Stray light emission (conventional types)	-		---	---	---	---
5226	Line width A	-		Width	---	0.50	mm
5226	Line width B	-		Width	---	0.75	mm
5223	Modulation	-		ΔEc	---	50	Vdc
5221	Light output	2		Light	4	---	ftL
(F) 5216	Cathode illumination	3		---	---	---	---
(F) 1002	Barometric pressure, reduced	3	54 ±5 mmHg	---	---	---	---
(F) 1141	Pressure	3		---	---	---	---
(F) 1331	Direct-interelectrode capacitance	3	k to all	Ck	---	2.5	pF
			g1 to all	Cg1	---	5	pF
			D1 to D2	C1D2	---	1.25	pF
			D3 to D4	C3D4	---	1	pF
			D1 to all	CD1	---	3	pF
			D2 to all	CD2	---	4	pF
			D3 to all	CD3	---	3	pF
			D4 to all	CD4	---	3	pF
(F) 5111	Vibration	3		Width	---	1.0	mm
(F) 5115	Shock	3		---	---	---	---
(F) 1121	Base strain	3		---	---	---	---

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. The light output shall be taken using a 1/2 by 1/2 inch (12.7 by 12.7 mm) raster of 35 to 105 lines and 60 frames per second. Ik = 500 µAdc.
3. (F) This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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NOTES: -Continued

4. First article inspection shall consist of performing all tests listed on this tube specification sheet. The sample size and allowable defects shall be in accordance with MIL-E-1, Qualification. Samples and acceptance criteria. Three copies of the test report, prepared in accordance with MIL-STD-831, shall be forwarded to the purchasing activity for evaluation by the preparing activity. Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

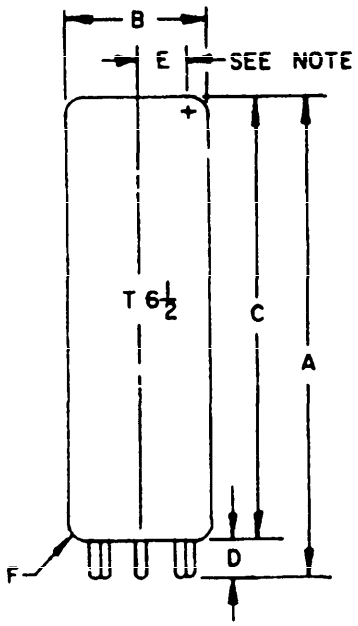
User activities:
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Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
F	Base: E9-1 except for pin length	
Quality conformance inspection, part 2		
A	2.917(74.09)	3.004(76.30)
B	.750(19.05)	.875(22.23)
C	2.687(68.25)	2.749(69.82)
D	.230 (5.84)	.255 (6.48)
E	.350 (8.89)	- - -

Pin No.	Element
1	Grid No. 1
2	Deflector plate D1
3	Deflector plate D3
4	Heater
5	Heater cathode
6	Deflector plate D2
7	Anode No. 2
8	Anode No. 1
9	Deflector plate D4

NOTE: Useful screen radius.

FIGURE 1. Outline drawing of electron tube type 1DP1.

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